

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	20 June 2026
Team ID	24EG109A55 (Turlapati Lokesh)
Project Name	SHOPEZ - Online Shopping Application
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Product Catalog	Browse products by category Search products by name Filter by price / rating
FR-4	Cart & Order Management	Add / remove items from cart Place order & view order confirmation Track order status
FR-5	Admin / Seller Panel	Add, edit, delete products View and update order status Manage inventory / stock
FR-6	User Profile	View and edit profile details View order history

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple, intuitive UI so users can browse and check out with minimal clicks, on both desktop and mobile.

NFR-2	Security	User passwords are hashed and authentication is handled via JWT tokens; role-based access control restricts admin routes to authorized users only.
NFR-3	Reliability	The system should handle failed transactions gracefully and ensure cart/order data is not lost due to network or server errors.
NFR-4	Performance	Product listing and search APIs should respond within 1-2 seconds under normal load, using indexed MongoDB queries.
NFR-5	Availability	The application should be deployed on a reliable hosting platform (e.g. Render/Vercel) targeting close to 99% uptime.
NFR-6	Scalability	The REST API and MongoDB schema are designed so new modules (payments, reviews) can be added without major redesign, and the backend can scale horizontally.